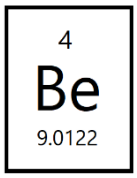
Science

2024/2025 – Middle Grades – Practice 2

Student Instructions:

- Write your complete ID code on your answer sheet.
- If you change your answer, erase well.

1. The diagram provided is a cell from the periodic table representing the element beryllium. From this information, how many neutrons would you expect an average beryllium atom to have?



   A. 4
   B. 5
   C. 9
   D. 13

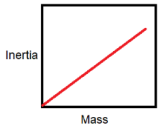
2. Consider the chemical reaction __Fe_2O_3 + __C → __Fe + __CO_2. Which of the following coefficients can be used to balance this chemical equation?
   A. 1, 1, 2, 3
   B. 2, 2, 3, 2
   C. 1, 3, 2, 1
   D. 2, 3, 4, 3

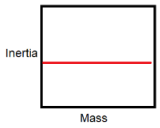
3. This law assumes that pressure remains constant for an ideal gas. What law explains that as a gas's temperature increases, so does its volume?
   A. Boyle's law
   B. ideal gas law
   C. Charles's law
   D. Avogadro's law

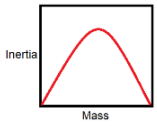
4. This form of matter was finally produced in 1995 when a small gaseous sample of rubidium atoms was supercooled to almost absolute zero. What fifth form of matter is this?
   A. Bose-Einstein condensate
   B. plasma
   C. superfluid
   D. cryolite

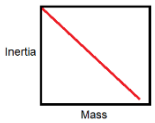
5. Visible light and other forms of electromagnetic radiation take the form of this type of wave. What type of wave does not need a medium through which to travel, and therefore can travel through the vacuum of space?
   A. longitudinal waves
   B. transverse waves
   C. de Broglie waves
   D. compressional waves

6. Which of the following graphs best describes the relationship between mass and inertia of an object?

A. 

B. 

C. 

D. 

7. The photo provided shows a roller coaster going through an inverted loop. What force shown by the arrow is the force that pulls an object towards the center of a circle around which the object is moving?

A. centrifugal force
B. normal force
C. tangent force
D. centripetal force

8. A 2 ampere current passes through a 10 ohm resistor in a simple electric circuit. What is the voltage in volts of this circuit?
A. 0.2 V
B. 40 V
C. 20 V
D. 5 V

9. All forms of electromagnetic radiation are characterized by their wavelength. Which of the following forms of electromagnetic radiation possesses the longest wavelength?
A. infrared waves
B. microwaves
C. ultraviolet rays
D. visible light

10. A 0.1 kilogram baseball is traveling at 10 meters per second. What is the object's kinetic energy in joules as it is moving?
A. 5 J
B. 25 J
C. 1 J
D. 10 J

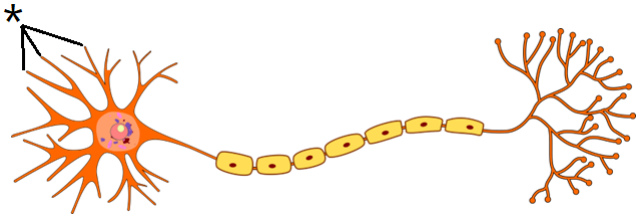
11. Zero Kelvin is considered to be Absolute Zero. To the nearest degree, what is the equivalent of 0 Kelvin in degrees Celsius?
A. 0°C
B. 32°C
C. -100°C
D. -273°C

12. Match the cell organelle with its primary function:

x. mitochondrion	1. produces ATP energy for the cell
y. chloroplast	2. breaks down cellular materials
z. lysosome	3. produces sugars, such as glucose, for the cell

A. x-1, y-2, z-3
B. x-1, y-3, z-2
C. x-2, y-1, z-3
D. x-3, y-2, z-1

13. These structures of the neuron receive stimuli and generate electrochemical messages that travel down the neuron. What structures are shown by the asterisk (*) in the diagram provided?



A. soma
B. myelin sheathes
C. dendrites
D. axons

14. Which of the following unicellular organisms are considered to be prokaryotic?
A. dinoflagellates
B. amoebas
C. protozoa
D. bacteria

15. In an adult human, these organs are about 8 to 10 inches long. What pair of tube-like organs transport urine from the kidneys to the urinary bladder?
A. carotid canals
B. vas deferens
C. urethrae
D. ureters

16. Match the type of RNA with its description:

x. mRNA	1. carries information from the nucleus to the ribosome
y. rRNA	2. combines with protein to form ribosomes
z. tRNA	3. carries amino acids to the ribosome

A. x-1, y-2, z-3
B. x-2, y-3, z-1
C. x-3, y-1, z-2
D. x-3, y-2, z-1

17. Meiosis produces this type of cell. What term refers to sex cells such as egg and sperm?
   A. prezygote
   B. gamete
   C. zygote
   D. spore

18. The X- and Y-chromosomes are considered to be the sex chromosomes because they determine what biological sex an individual will be. Which combination of these chromosomes results in a male in humans?
   A. YY
   B. XX
   C. XXX
   D. XY

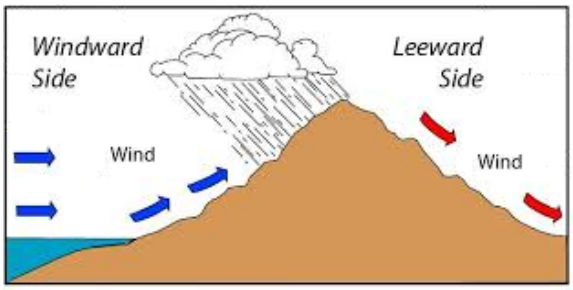
19. Observe the ocean food chain below. Which organism is classified as a producer?

   A. tuna
   B. phytoplankton
   C. orcas
   D. zooplankton

20. This biome is typically treeless and experiences short summers. Which of the following terrestrial biomes is known for its permafrost?
   A. savanna
   B. tundra
   C. temperate grassland
   D. taiga

21. Which of the following best represents a biological community?
   A. antelopes, zebras, soil, tropical grasses, termites, water, rocks, and cheetahs
   B. antelopes
   C. antelopes, zebras, tropical grasses, termites, and cheetahs
   D. antelopes, polar bears, tropical grasses, nurse sharks, ants

22. The diagram provided shows how winds from the ocean lose their water content through precipitation when they increase elevation and pass over mountains. The resulting winds on the leeward side of the mountain range tend to have very low water content and deserts often form in that area. What phenomenon is shown by this diagram?



   A. drought
   B. anticyclone
   C. monsoon
   D. rain shadow

23. Metamorphic rocks form under high heat and pressure deep within the Earth. Which of the following rocks is considered to be metamorphic?
   A. marble
   B. basalt
   C. granite
   D. shale

24. The atmosphere is a mixture of different gases. What is the most abundant gas in our atmosphere?
    A. CO_2
    B. N_2
    C. O_2
    D. O_3

25. This type of tide results when the moon and sun are at right angles to each other. Which type of tide represents a period in which the difference between the high tide and low tide is the lowest value?
    A. spring tide
    B. ebb tide
    C. neap tide
    D. diurnal tide

26. Seismic waves change their velocity when passing this boundary. What is the region that is the boundary between the crust and mantle?
    A. Mohorovicic discontinuity
    B. Karman line
    C. stratopause
    D. Lehmann discontinuity

27. These deep space objects are powered by supermassive black holes. What are the extremely hot and luminous centers of active galaxies?
    A. pulsars
    B. supernovae
    C. planetary nebulas
    D. quasars

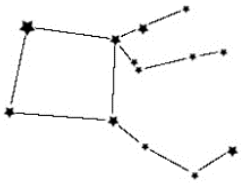
28. This object was discovered by Giuseppe Piazzi in 1801. What dwarf planet is located within the asteroid belt?
    A. Ceres
    B. Sedna
    C. Mimas
    D. Charon

29. Match the American astronaut with his major accomplishment:

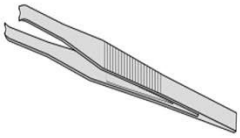
x. John Glenn	1. 1st American to complete a spacewalk
y. Alan Shepard	2. 1st American in space
z. Ed White	3. 1st American to orbit the Earth

    A. x-1, y-3, z-2
    B. x-3, y-2, z-1
    C. x-2, y-1, z-3
    D. x-2, y-3, z-1

30. Observe the constellation provided. What is the name of this constellation?
    A. Sagittarius
    B. Pegasus
    C. Draco
    D. Sirius major



31. These tools resemble tweezers and they are used to grasp and manipulate soft tissues in a biological or medical setting. What scientific tool is shown in the picture to the right?



A. calipers
B. manometer
C. tongs
D. forceps

32. A student is interested in determining if the color of light affects the growth of plants. She sets up an experiment involving two groups of plants. Group 1 plants are placed under a red light and Group 2 plants are placed under a white light that mimics the wavelengths of sunlight. Both groups of plants are measured daily and cared for equally throughout the experiment. Which of the following would represent the "control" of the experiment?
A. The control of the experiment would be the plants of Group 2 because they are used as a standard of comparison.
B. The color of light used in each experiment is the control because they are variables.
C. The equal care given to both groups are a control because they both need the same water, nutrients, and other factors.
D. The growth of both groups of plants act as a control because it involves the results of the experiment.

33. The scientific method requires a hypothesis to be formed when attempting to answer a question about a phenomenon. Which of the following statements best defines an experimental hypothesis?
A. A hypothesis makes a conclusion based on collected data from an experiment.
B. A hypothesis is a possible explanation that can be tested by an experiment.
C. A hypothesis is any educated guess.
D. A hypothesis is an observation of a scientific phenomenon.

34. These organisms lack both xylem and phloem. Which of the following are classified as nonvascular plants?
A. flowering plants
B. conifers
C. ferns
D. mosses

35. It is thought that plants diverged from these organisms about 500 million years ago. Which organisms are thought to be the closest ancestors to modern plants?
A. green algae
B. yeasts
C. diatoms
D. viruses

36. This structure is often green in color, but some species have some that resemble petals. What is the collection of a plant's sepals called?
A. corolla
B. calyx
C. anther
D. petiole

37. This phenomenon can be either positive or negative. What phenomenon occurs when a plant grows in response to physical touch?
A. haptotropism
B. gravitropism
C. phototropism
D. thigmotropism

38. Recent genetic evidence suggests that modern humans did not evolve from Neanderthals, but rather evolved alongside them from a common ancestor. What is the scientific name of modern humans that translates as "wise human"?
A. *Homo habilis*
B. *Australopithecus afarensis*
C. *Homo sapiens*
D. *Homo neanderthalensis*

39. Ivan Pavlov discovered this behavior when he noted that dogs would salivate whenever he rang a bell as he was feeding them. What is this behavior in which learning occurs through the association between two unrelated events?
A. imitation
B. instinctive learning
C. imprinting
D. classical conditioning

40. Match the insect with its correct taxonomic order:

x. orthoptera	1. bees, wasps, ants
y. coleoptera	2. grasshoppers, katydids, crickets
z. hymenoptera	3. beetles

A. x-1, y-2, z-3
B. x-2, y-1, z-3
C. x-2, y-3, z-1
D. x-3, y-1, z-2

41. This disease is caused by a bacterium in the genus *Clostridium*. Which of the following diseases is a type of food poisoning resulting from an infection of bacteria?
A. meningitis
B. measles
C. botulism
D. malaria

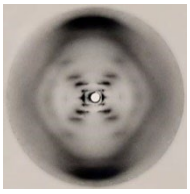
42. A lysogenic viral life cycle has this stage. What is a dormant virus that hides in its host DNA for several generations of cell divisions?
A. prophage
B. cryptid
C. endospore
D. viroid

43. What is the study of amphibians and reptiles?
A. mycology
B. ichthyology
C. herpetology
D. entomology

44. This system uses everyday physical objects that have microprocessors embedded within them. What term refers to a network of "smart objects" that are connected to each other and the internet that can exchange data with other devices or systems?
A. V2X communications
B. cloud computing
C. The Internet of Things
D. blockchain

45. This aviator is credited with developing the first practical American helicopter in 1939. Who designed the *R-4* from that original model, creating the world's first mass-produced helicopter?
A. Igor Sikorsky
B. Orville Wright
C. Charles Lindbergh
D. Henri Coanda

46. Photo 51, shown here, provided valuable information that helped James Watson and Francis Crick determine the double helix structure of DNA. What female biochemist is credited with taking Photo 51 along with her assistant Raymond Gosling?



A. Irene Curie
B. Jennifer Doudna
C. Rosalind Franklin
D. Dorothy Hodgkin

47. Match the blood vessel with its function:

x. artery	1. carries blood towards the heart
y. vein	2. site of gas and nutrient exchange
z. capillary	3. carries blood away from the heart

A. x-1, y-3, z-2
B. x-2, y-3, z-1
C. x-3, y-1, z-2
D. x-3, y-2, z-1

48. Organic nutrients provide energy when consumed. Which basic nutrient below provides the most energy in Calories per gram?
A. lipids
B. minerals
C. proteins
D. carbohydrates

49. This condition has symptoms that include weakness, fatigue, sore arms and legs, irritability, joint pain, and bleeding gums. What condition is a result of vitamin C deficiency?
A. scurvy
B. beriberi
C. rickets
D. pellagra

50. Hemoglobin is a protein complex found in red blood cells that carries oxygen from the lungs to the tissues. What metal ion is essential to the "heme" portion of hemoglobin?
A. copper
B. calcium
C. magnesium
D. iron

Science – Middle Grades – Practice 2
2024-25 Governor's Cup

1. B
2. D
3. C
4. A
5. B
6. A
7. D
8. C
9. B
10. A
11. D
12. B
13. C
14. D
15. D
16. A
17. B
18. D
19. B
20. B
21. C
22. D
23. A
24. B
25. C
26. A
27. D
28. A
29. B
30. B
31. D
32. A
33. B
34. D
35. A
36. B
37. D
38. C
39. D
40. C
41. C
42. A
43. C
44. C
45. A
46. C
47. C
48. A
49. A
50. D